

ABROR RAHMATULLAH

Backend Engineer | Full-Stack Developer | Junior AI Engineer

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PROFESSIONAL SUMMARY

Full-stack and AI engineer with 4+ years of experience delivering production-grade Python systems across fintech, logistics, and enterprise environments. Specializes in REST API development (FastAPI, Django, Flask), multi-database architecture (PostgreSQL, SQL Server, MongoDB, MySQL), and building scalable data platforms.

Proven track record in developing centralized data systems (Single Source of Truth), including a Flask-based enterprise platform with integrated Excel ingestion pipelines, audit trail logging, and production deployment with robust error handling—improving data consistency, traceability, and operational efficiency across multiple divisions.

Distinctively experienced in building AI-integrated applications for data-sensitive and regulated environments, including an offline RAG chatbot with local LLM inference (Ollama, DeepSeek, FAISS) deployed within a secure government intranet, ensuring zero external data exposure.

Additional highlights include engineering an asynchronous iDeb SLIK credit report pipeline for an Indonesian state-owned financial institution, as well as developing Odoo ERP integrations, Order Management Systems, and cross-platform APIs supporting real-world business operations.

Also holds a Golang Backend Development certification with hands-on production exposure. Actively seeking remote backend or AI engineering roles with global companies.

TECHNICAL SKILLS

Languages: Python (advanced) · PHP · Go (Golang) · JavaScript · SQL

Backend Frameworks: FastAPI · Django · Django REST Framework · Flask · Laravel

AI / ML: Ollama · DeepSeek LLM · FAISS · HuggingFace Embeddings · RAG · LangChain · Semantic Search

Databases: PostgreSQL · Microsoft SQL Server · MySQL · MongoDB · Redis

API & Architecture: REST API Design · JSON API · Odoo XML-RPC API · Async APIs · Microservices · ORM

Tools & DevOps: Git · GitHub · Docker (basic) · Linux · Postman · asyncio · Pandas · Base64 · queue.Queue

Front-end: HTML5 · CSS3 · JavaScript (ES6+) · Bootstrap

PROFESSIONAL EXPERIENCE

PT Sarana Multi Infrastruktur (Persero)

Full-Stack Python Developer & AI Engineer • Jakarta, Indonesia (Hybrid)

Jan 2025 – Present

State-owned infrastructure finance company — building AI-integrated backend systems in a regulated financial environment where data privacy and system reliability are non-negotiable.

- Engineered a high-throughput iDeb SLIK credit report processing pipeline (Flask, asyncio, queue.Queue, threading) handling large Base64-encoded file uploads asynchronously — eliminating main thread blocking and reducing processing latency for financial data workflows.
- Built and deployed a fully offline RAG-based AI chatbot (Ollama + DeepSeek LLM + FAISS + HuggingFace multilingual embeddings + Flask) for secure document Q&A inside a government intranet — designed for environments where no data can leave the internal network.

- Implemented FAISS vector store with multilingual sentence embeddings enabling semantic search across institutional documents, replacing keyword-only retrieval and dramatically improving staff information-finding speed.
- Integrated internal systems with Odoo ERP via REST API (Python), automating finance operations data sync and eliminating manual cross-system entry for the operations team.
- Designed SQL Server schemas and MongoDB collections for chat history, source document metadata, and financial record storage, optimizing queries for concurrent multi-user access.
- Developed a centralized Single Source of Truth web application using Flask and SQLAlchemy, enabling multiple divisions to access and manage consistent data from a single, reliable platform.
- Integrated an Excel file ingestion pipeline using Pandas and openpyxl, allowing users to upload and process structured data directly into SQL Server via pyodbc and SQLAlchemy.
- Designed and implemented an Audit Trail module to log all user activity, ensuring full data accountability and traceability across the platform.
- Deployed the application to production using Waitress WSGI server with centralized logging and robust error handling for 404, 500, and unhandled exceptions.
- Stack: Python · Flask · FastAPI · asyncio · Pandas · Ollama · DeepSeek · FAISS · HuggingFace · SQL Server · MongoDB · Odoo API

PT Lintas Samudera Borneo Line

IT Developer Officer • Jakarta, Indonesia

Aug 2022 – Jan 2025 (2 yrs 6 mos)

National logistics and shipping company — owned the full software development lifecycle for internal systems across order management, ERP integration, and operations.

- Designed and built REST APIs for Odoo ERP integration using Python (FastAPI, Django REST Framework) and MySQL, enabling automated data sync between logistics operations and finance systems.
- Re-engineered the legacy VVIS platform into a modern Operations Management System using Python (Django) and PostgreSQL — improving system reliability, query performance, and operational visibility across the company.
- Developed and deployed Phase 2 of the Order Management Application (PHP, Laravel, HTML, MySQL), streamlining data entry workflows and improving UX for operations staff across multiple branches.
- Maintained and stabilized existing PHP/HTML/MySQL applications serving daily logistics operations, reducing downtime and resolving accumulated technical debt.
- Stack: Python · Django · FastAPI · Django REST Framework · PHP · Laravel · MySQL · PostgreSQL · Odoo API · HTML · CSS · JavaScript

PT Multipolar Technology Tbk

Application Developer • Tangerang, Indonesia

Jun 2021 – Jun 2022 (1 yr)

Technology consulting firm — built and extended APIs for enterprise client systems.

- Developed REST APIs for new and existing client systems using Python (FastAPI) and MongoDB, enabling integration with third-party services and internal platforms.
- Analyzed business requirements and authored software specifications, translating complex client needs into implementable engineering designs.
- Conducted pre-QA testing of new features and enhancements, catching bugs before formal quality assurance review.
- Stack: Python · FastAPI · MongoDB · REST API Design

PT Industri Kereta Api (INKA) — Persero

IT Staff Intern • Madiun, Indonesia

Jan 2019 – Feb 2019 (2 mos)

- Developed employee data management features (CRUD) for the APPS IT information system using PHP and Laravel, used by HR operations.
- Conducted functional testing on newly developed application features and documented findings for the development team.
- Stack: PHP · Laravel · MySQL

KEY PROJECTS

Centralized Data Platform (Single Source of Truth) — Python · Flask · SQLAlchemy · Pandas · SQL Server · pyodbc · openpyxl · Waitress

- Problem: Multiple divisions relied on fragmented data sources, causing inconsistencies, duplication, and lack of data accountability.
- Solution: Designed and developed a centralized web-based platform using Flask and SQLAlchemy to serve as a Single Source of Truth, enabling consistent data access and management across divisions.
- Built an Excel ingestion pipeline using Pandas and openpyxl, allowing users to upload and process structured data directly into SQL Server via pyodbc and SQLAlchemy.
- Implemented a comprehensive Audit Trail system to log all user activities, ensuring full data traceability, compliance, and accountability.
- Deployed the application to production using Waitress WSGI, with centralized logging and robust error handling (404, 500, and global exception handling) to improve system reliability.
- Impact: Improved data consistency across teams, reduced manual processing, and increased transparency and auditability of critical business data.

iDeb SLIK Credit Report Processing Pipeline — Python · Flask · asyncio · Pandas · SQL Server

- Problem: Large Base64-encoded credit report files caused blocking and timeouts in the synchronous upload flow.
- Solution: Redesigned as an async pipeline using asyncio, queue.Queue, and threading — decoupling upload from processing, converting JSON to structured tables via Pandas, storing metadata in SQL Server.
- Impact: Eliminated main-thread blocking, enabling concurrent file handling and significantly reducing processing latency for credit assessment workflows.

Operations Management System (VVIS Re-engineering) — Python · Django · PostgreSQL · REST API

- Migrated a legacy operations platform to a modern Django + PostgreSQL architecture, improving data integrity, system performance, and maintainability for a national logistics company.
- Designed and implemented Odoo ERP API integrations alongside the migration, enabling automated financial data sync between operations and ERP systems.

EDUCATION

Politeknik Negeri Malang

2016 – 2020

Bachelor's in Applied Computer Science · GPA: 3.62 / 4.00 · TOEIC: 665 (B2 English proficiency)

CERTIFICATIONS & TRAINING

Odoo Development Training — Arkana Solusi Digital (Feb 2024)

Golang Backend Development Certificate — SanberCode (Aug 2023) · JSON · REST APIs · Git · Go

+ 6 additional certifications in software development and AI